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Confined Cell Migration and Asymmetric Hydraulic Environments to Evaluate the Metastatic Potential of Cancer Cells

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Metastasis, a hallmark of cancer development, is also the leading reason for most cancer-related deaths. Furthermore, cancer cells are highly adaptable to micro-environments and can migrate along pre-existing channel-like tracks of anatomical structures. However, more representative three-dimensional models are required to reproduce the heterogeneity of metastatic cell migration in vivo to further understand the metastasis mechanism and develop novel therapeutic strategies against it. Here, we designed and fabricated different microfluidic-based devices that recreate confined migration and diverse environments with asymmetric hydraulic resistances. Our results show different migratory potential between metastatic and nonmetastatic cancer cells in confined environments. Moreover, although nonmetastatic cells have not been tested against barotaxis due to their low migration capacity, metastatic cells present an enhanced preference to migrate through the lowest resistance path, being sensitive to barotaxis. This device, approaching the study of metastasis capability based on confined cell migration and barotactic cell decisions, may pave the way for the implementation of such technology to determine and screen the metastatic potential of certain cancer cells.
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Introduction

Invasion and metastasis are the most dangerous hallmarks of cancer development [1,2], a multistage process where cancer cells spread and colonize distant organs. Unfortunately, despite remarkable progress in the field of cancer research, it still represents one of the leading reasons for most cancer-related deaths [3,4]. Metastatic cells have acquired the ability to invade the surrounding tissues, leading to the formation of secondary tumors. This journey entails a series of multistep stages. It commonly comprises cell dissociation from the primary tumor and epithelial-to-mesenchymal transition [5]. Later, motile cells usually go through different physical barriers, such as the basement membrane and the extracellular matrix (ECM). This drive ends with intravasation to the microvasculature of the lymph and blood systems as one of the last steps before motile cancer cells extravasate and spread to new locations [6,7].

In vivo, motile cancer cells undergoing metastasis are highly adaptable to the physicochemical properties of a wide variety of micro-environments. Generally, surrounding tissue invasion involves cell migration along pre-existing channel-like tracks of anatomical structures or by degrading the extracellular matrix if no pathway is available [8–10]. Furthermore, mechanosensing feedback and cytoskeletal rearrangement for drastic changes in cell shape are essential for invasive cancer cells in heterogeneous micro-environments, which usually include confined and constricting spaces narrower than the cell diameter [11,12].

In the last few decades, we have witnessed an extraordinary advance in cancer research as a result of a vast effort in the scientific community, and recently, several molecular markers have been successfully related to the metastatic potential of certain tumors [13]. However, the prediction of the metastatic potential of certain tumors still represents an unsolved challenge. Classical models of cell migration have typically been studied in two dimensions [14,15]. Nevertheless, those models do not accurately represent the real three-dimensional (3D) micro-environment present under in vivo cell migration. Thus, more representative 3D models are required to reproduce the heterogeneity of metastatic cell migration in vivo [16,17]. To overcome this limitation, microfluidic devices are helpful tools [18]. They can be used to recreate different cell migration-inducing mechanisms that rely on biochemical, mechanical, and topographical properties of the surrounding environment [19–22]. At the same time, a better understanding of the mechanisms by which cells migrate may lead to the development of novel therapeutic strategies for controlling tumor invasiveness and metastasis, mainly known as migrastatic drugs [23].

One of these mechanisms that regulate cell migration is barotaxis [24]. It mainly arises when confined cells move through low-permeability environments [24,25]. Under these conditions, cells have to displace the surrounding fluid to migrate, thus generating a pressure drop between the front of the cell and the end of its path. This pressure drop is proportional to the column of fluid the cell has to displace, namely, hydraulic resistance [24,26]. Such resistance is known to play a critical role in cell orientation, polarization, and migration itself [27]. Prentice-Mott et al. [28], pioneers in this field, revealed that leukocyte-like cells surrounded by asymmetric hydraulic environments are able to identify the path that poses the lowest hydraulic resistance. They observed that in certain cell types, the physical properties of the surrounding

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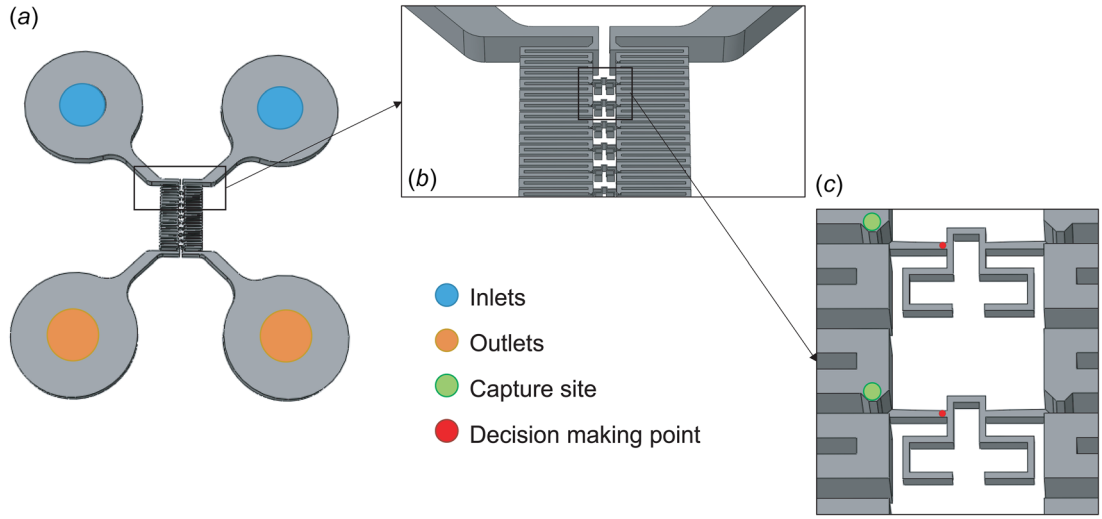


Fig. 1 Three-dimensional overview and details of one of the designed devices. (*a*) Full view in which 40-μm height inlets (top/blue dots) and outlets (bottom/orange dots) are connected through the core part of the device. (*b*) Magnification of the core serpentine paths of 40-μm height and the specific internal geometry of 10-μm height. (*c*) Details of the 10-μm height capture site (bigger/green dots) and the 10-μm height internal confined geometry (dead-end path in this case). Here, a short and confined channel connects the capture site with the decision-making point (smaller/red dots).

environment alone affect the decision-making process of directional cell migration and can even outcompete the impact of chemical properties [28,29]. Notwithstanding, this mechanism has been poorly studied in cancer cells.

The goal of this work was to develop microfluidic devices with confined migration channels and bifurcating pathways of different hydraulic resistances. These devices aim to shed light on cancer cell migration and barotaxis and thus to help screen metastatic cells and decipher the mechanism beyond the capability of certain tumor cells to undergo metastasis. Specifically, we designed and fabricated (I) devices that recreate confined migration and two paths of asymmetric hydraulic resistance; (II) negative-control devices, with symmetric paths; and (III) positive-control devices, with a dead-end in one of their paths. Furthermore, we fabricated (IV) devices with similar hydraulic resistances but different levels of tortuosity, to determine if the capacity of the cell to sense different hydraulic resistances could be biased by the topology of the device.

Therefore, to study cancer cell migration in confined environments and to determine if this migration is influenced by barotaxis, we used two different common cell lines of breast cancer, MCF-7, which is well known as nonmetastatic, and the MDA-MB-231 cell line, which has a high invasive capability [30].

Methods

Design of Microfluidic Devices for Single-Cell Migration Under Barotactic Stimuli. To better understand how cancer cell migration is affected by confined micro-environments and barotactic stimuli, novel microfluidic devices were fabricated, and tumor cells were left to choose between two paths whose main difference was their hydraulic resistance. These devices were designed with two different heights and have two main functional parts (Fig. 1): a cell capture site and a decision-making point, as explained later.

The first part of the designed devices consists of a single-cell capture site (Fig. 1(*c*)) created by a subcellular size hole with lower hydraulic resistance [31–36]. These microfluidic chips allow single-cell hydrodynamic capture when cells are loaded through the inlet ports. Capture sites were built with a serpentine path of 40-μm height and 30-μm width and small gaps between its walls (of 10-μm height and 10-μm width) adjacent to the confined channels to improve cell capture and single-cell entry to the confined microchannels. The second part of the designed device consists of a channel (10 μm high and 8 μm wide, with a slightly enlarged entrance that was 10 μm high and 10 μm wide) that leads to the decision point, bifurcated into two channels of identical section (10 μm high and 8 μm wide). This channel is located in the middle of the internal geometry, where the cell must migrate through.

Once cells have been trapped at the cell capture site (Fig. 1(*c*)), they can take several actions: they can either continue through the central channel, migrate through the serpentine, or enter into the confined migration channel in which they will have to choose once in the bifurcation point. At this point, the only difference between the two paths is the hydraulic resistance the cell senses when trying to enter each of the paths [28].

Four internal geometries (Fig. 2) were designed: (I) a symmetric device (Fig. 2(*a*)), used as a negative control, where the decision paths (top path and bottom path) are identical, as well as their hydraulic resistances, and the cells are expected to adopt a random choice; (II) a twisted device (Fig. 2(*b*)) in which one of its paths is twisted and therefore has a higher length and higher hydraulic resistance (long path and short path); (III) a dead-end device (Fig. 2(*c*)), exhibiting one path where there is no exit (dead-end path and open path) to maximize the hydraulic-resistance difference between both paths (as the only outflow is occluded by the presence of the cell) [37]; and finally, (IV) a tortuosity device (Fig. 2(*d*)), where the two decision paths present similar lengths but different levels of tortuosity (high-tortuosity path and low-tortuosity path).

Polydimethylsiloxane-Based Microfluidic Devices. Polydimethylsiloxane (PDMS)-based microfluidic devices were fabricated as described by Shin et al. [38]. Briefly, soft lithography was used to develop two aligned heights of positive SU8 molds onto silicon wafers with the desired geometry (micro-LIQUID, Arrasate/Mondragón, Spain). Microdevices were fabricated in PDMS (Sylargd-184, Dow, Midland, TX) at a 10:1 weight ratio for the base to curing agent. The solution was mixed and poured onto the SU8 master and then degassed to remove air bubbles. PDMS was cured at 80 °C for 24 h, and then, the replica-molded layer was trimmed, perforated, and autoclaved. PDMS microdevices were activated with plasma treatment and bonded to 35-mm glass-bottom petri dishes (Ibidi, Gräfelfing, Germany).

Collagen type I (20 μg/mL) was coated for 1 h at 37 °C to facilitate cell adhesion. After incubation, the microdevices were

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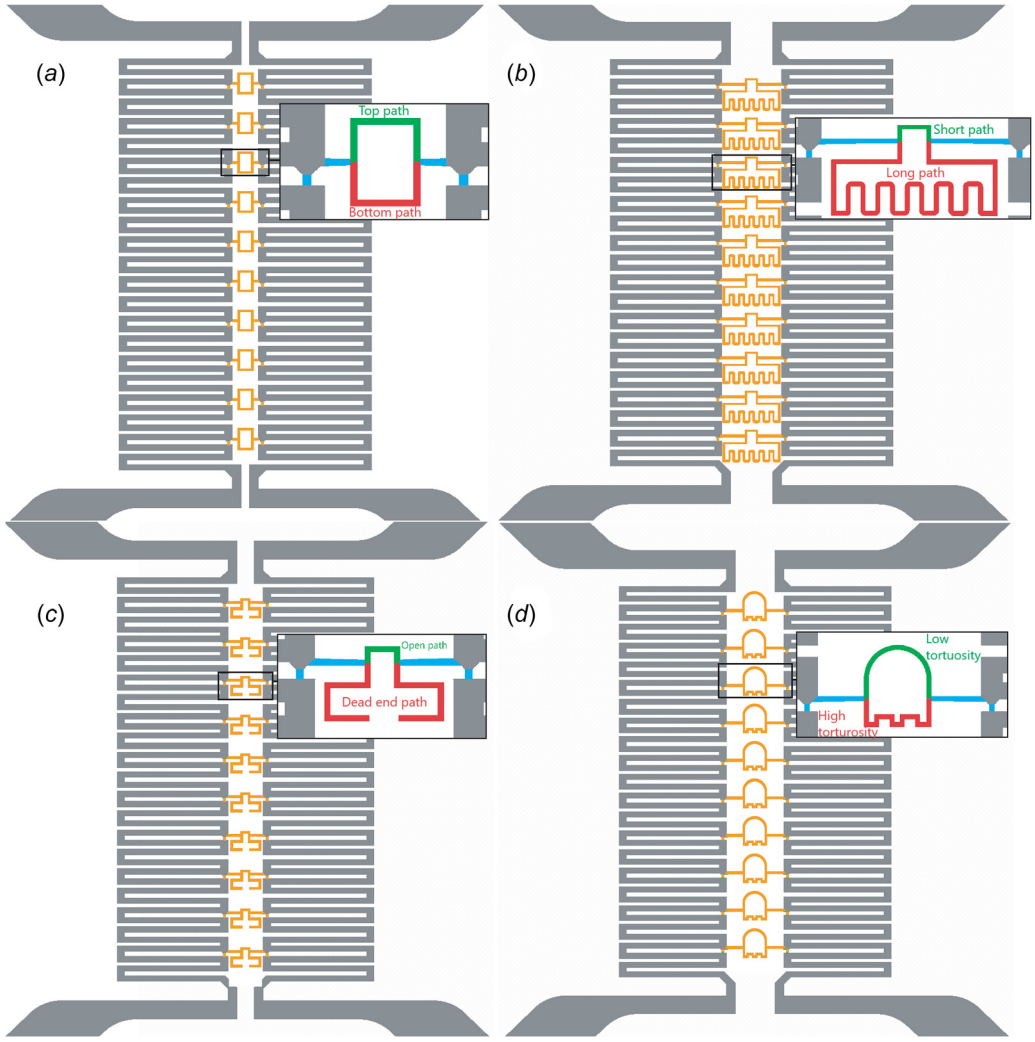


Fig. 2 Scheme of the designed devices: gray color (external part) identifying 40-μm height structures, and orange color (central/internal part) representing 10 μm. Symmetric device (*a*), twisted device (*b*), dead-end device (*c*), and tortuosity device (*d*). The zoom at the central part of the device shows the possible migration paths for the barotactic study: top/green and bottom/red.

cleaned with phosphate-buffered saline (Gibco, Thermo Fisher Scientific, Waltham, MA) and Dulbecco's modified Eagle's medium (DMEM, Gibco). Once the device was fully primed with culture media, cells resuspended at a concentration of 1.5×10^6 cells/ml of complete media were loaded *via a* static head (70 μL) onto the left inlet with the same volume of complete media in the right inlet (Fig. 1(*a*)). Cells were allowed to flow by the serpentine path (Fig. 1(*b*)), and after they were trapped in the capture sites (Fig. 1(*c*)), the fluid flow was stopped by adding complete media on the outlets. To avoid pressure drops between inlets, outlets, and the counterpart serpentine paths that could interfere with the cell performance, microdevices were allowed to settle for 30 min at 37 °C before image acquisition. Therefore, the device reaches static conditions, and the behavior of the cells during the experiment was exclusively based on the features of the channels. Hence, barotaxis can be measured, as the confined migration of the cells requires them to generate pressure to push the fluid ahead. The minimal pressure required for cell migration in one channel is proportional to the hydraulic resistance of that channel (defined by its length and tortuosity, being the cross section constant among all the channels) [24,26].

Image Acquisition and Analysis. Cell migration was visualized and recorded *via* time-lapse live microscopy (Nikon-Eclipse Ti, Nikon, Tokyo, Japan) at 37 °C in a humidified atmosphere of 5% CO_2. The microscope was equipped with a cell culture chamber, which maintained ideal temperature, humidity, and O_2/CO_2 conditions for cell culture. Phase-contrast time-lapse images were taken for 18 h in a 5-min interval and using a 10× objective. Only single and viable captured cells were tracked over time. Manual counting following the acquired images was performed to assess confined channel internalization and to evaluate migration path decisions among those cells that reached the confined bifurcation point. Between 4 and 8 devices of each different geometry were tested per cell line. In addition, migration speed measures to calibrate fluid flow simulations were carried out *via* the ImageJ manual tracking plug-in (National Institutes of Health, USA).

Quantification of the Hydraulic Resistance. To understand the barotactic stimulus that cells sense at the decision-making point (Fig. 1), we evaluated the hydraulic resistance of the channel by two different methods. On the one hand, we computationally simulated the two-dimensional steady fluid flow to study the difference in the hydraulic resistance between the paths of each microdevice. On the other hand, theoretical approximations of the hydraulic resistance of some straight channels were estimated following previous theoretical works [26]. However, with this theoretical approximation, the hydraulic losses produced in elbow pipping are underestimated. In the case of the tortuosity microdevice, since the lengths of the two channels are similar, the losses

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produced in the change in direction of the pipping could modify the expected hydraulic resistance, affecting barotaxis. A comparison between the two methods is included in Supplemental Material on the ASME Digital Collection.

To better understand the hydraulic resistances at the decision-making point, we performed fluid flow simulations of the microdevices by means of the finite volume method in ANSYS FLUENT 2019 R2. The details of the fluid flow simulations have been incorporated in Supplemental Material. Moreover, to validate the computational fluid flow simulations and to guarantee the independence of the mesh on the solution, we performed a mesh convergence analysis by means of the grid convergence index presented by Roache [39] (Supplemental Material on the ASME Digital Collection).

Cell Lines. Human breast cancer cell lines were chosen since breast cancer is a major malignancy among cancers; it heads the number of cancer-related deaths in women worldwide, and its incidence continues to rise annually [40,41]. MCF-7 and MDA-MB-231 cells were purchased from the European collection of authenticated cell cultures as models of nonmetastatic and metastatic tumors. The cells were cultured at 37 °C in a humidified atmosphere of 5% CO_2. They were routinely grown in DMEM-high glucose (Gibco) supplemented with 10% heat-inactivated fetal bovine serum (Gibco) and 1% antibiotic-antimycotic solution of penicillin, streptomycin, and amphotericin B (Gibco). Nearly 1000 cells of both types were tested in total, and more than 70 were used for the hydraulic-resistance cell decision. Single-cell size was measured in suspended cells immediately after trypsinization using NIS-ELEMENT software (Nikon).

Cell Migration Within Three-Dimensional Extracellular Matrix. Alternative microdevices [20] were fabricated to host ECMs, following the methodology in the Methods section and the remaining steps of the Shin et al. protocol for cell culture within hydrogels in microfluidic devices.

Collagen gels of 2.5 mg/mL concentration and *p*H 7.4 were created by mixing collagen type I (BD Bioscience, BD, Franklin Lakes, NJ), NaOH, Dulbecco's phosphate-buffered saline (Thermo Fisher Scientific), and culture medium with suspended cells (after trypsinization and centrifugation). MCF-7 and MDA-MB-231 cells were mixed with gel solutions at a final concentration of 2×10^5 cells/ml and pipetted into the gel cavities of the microdevice, as in previous reports [20,42–46].

Subsequent to filling the gel scaffold regions, the devices were placed in prepared humid chambers within a CO_2 incubator to allow the collagen to polymerize at 37 °C for 20 min. After polymerization, the gels were hydrated with DMEM and left in the incubator for 24 h beforehand to ensure the stabilization of the matrix and cell adhesion and conditioning. Then, MCF-7 or MDA-MB-231 cells within the 3D extracellular matrix were subjected to phase-contrast imaging every 20 min for 24 h *via* time-lapse live microscopy at 37 °C in a humidified atmosphere of 5% CO_2. The focal plane located in the middle of the device along the *z*-axis was selected, which was more than 100 μm away from the two-dimensional surfaces of the device, ensuring fully embedded cells were analyzed [20,42]. Cell trajectory acquisition was performed using a hand-coded semi-automatic MATLAB script described in previous studies [20,42–46]. These trajectories were used to extract the mean speed and effective velocity, defined as the average instantaneous speed including all time steps and speed from the initial to the final position, respectively.

Results

Microfluidic-Based Chips With Different Channel Configurations Allow the Achievement of a Diverse Variety of Barotactic Stimuli That Could Be Sensed by Tumor Cells. To explore how mechanical factors can regulate cancer cell migration under physical confinement, we designed four microfluidic devices: a symmetric device, a twisted device, a dead-end device, and a tortuosity device (Fig. 2). The double height design allowed both the capture of individual cells and the connection of the loading cell ports to the confined area, where cells are tested against different hydraulic resistances. These devices enabled the isolation of mechanical factors from chemical factors and allowed the study of barotaxis and hydraulic resistance of individual cancer cells by decision-making.

To control the hydraulic resistance of the different paths, we studied the contours of the velocity magnitude and the velocity profile of the microdevices (Fig. 3). The velocity magnitude was higher in the lower hydraulic resistance paths and, as the sections were the same, it was proportional to the flow. Overall, the velocity at the walls was equal to zero to guarantee the no-slip condition at the boundary layers and the maximum at the center point of the section, which results in a classic velocity parabolic profile. In the case of the symmetric device (Fig. 3(*a*)), since the length and tortuosity of the paths are the same, the velocity profiles in both paths were identical, which means that there was no different hydraulic resistance between them. In the twisted device (Fig. 3(*b*)), the velocity in the short path was significantly higher than that in the long path. In particular, the flow along the short path was approximately twelve times higher than that along the long path, which represented approximately 92.5% of the inlet flow. In this asymmetric configuration, a hydraulic resistance difference of 85% was achieved. In the case of the dead-end microdevice (Fig. 3(*c*)), the fluid flowed in the open path since the dead-end path was closed and did not allow any flow, maximizing the hydraulic resistance difference between the channels. Finally, in the tortuosity microdevice (Fig. 3(*d*)), the velocities in both paths were very similar. The inlet flow was divided into 48.75% toward the low-tortuosity path and 51.25% toward the high-tortuosity path; therefore, the hydraulic resistance of the high-tortuosity path was 2.5% lower than that of the low-tortuosity path (Table S1 available in the Supplemental Materials).

MDA-MB-231 Metastatic Cells Tend to Migrate Through Paths of Lower Hydraulic Resistance. After the computational study of the microfluidic properties of the devices, different human breast adenocarcinoma cells (nonmetastatic MCF-7 and metastatic MDA-MB-231) were seeded in the devices. Once the cells were captured by the cell-single traps, the microfluidic chips were equilibrated and allowed to settle before time-lapse imaging.

Our results show that despite the nonmetastatic features of MCF-7 cells, they were surprisingly not able to reach decision-making sites (Video S1 available in the Supplemental Materials) in any of the internal geometries. They did not seem to have the ability to migrate in the confined microchannel (10 μm high and 8 μm wide). However, this was not observed for the metastatic MDA-MB-231 cells, which entered and moved along the confined microchannels in an apparently easy manner.

To assess whether hydraulic resistance affects the decision-making of individual cancer cells, the MDA-MB-231 cells were allowed to migrate within the different designed microdevices.

In the symmetric device (Fig. 4), which offered the same hydraulic resistance on both paths, MDA-MB-231 cells seemed to show no preference path of migration (45 versus 55%, devices 7, cells 18), as expected. However, we observed a marked decrease in cell migration toward high hydraulic resistance paths when using asymmetric devices (Twisted device) compared to the path with shorter length and therefore lower resistance (65% of them preferred the lower resistance channel, devices 8, cells 26) (Fig. 4).

To investigate the effect of barotaxis in metastatic cells more thoroughly, we developed a design that presented a dead-end path. In this scenario, the path that leads to the dead end would

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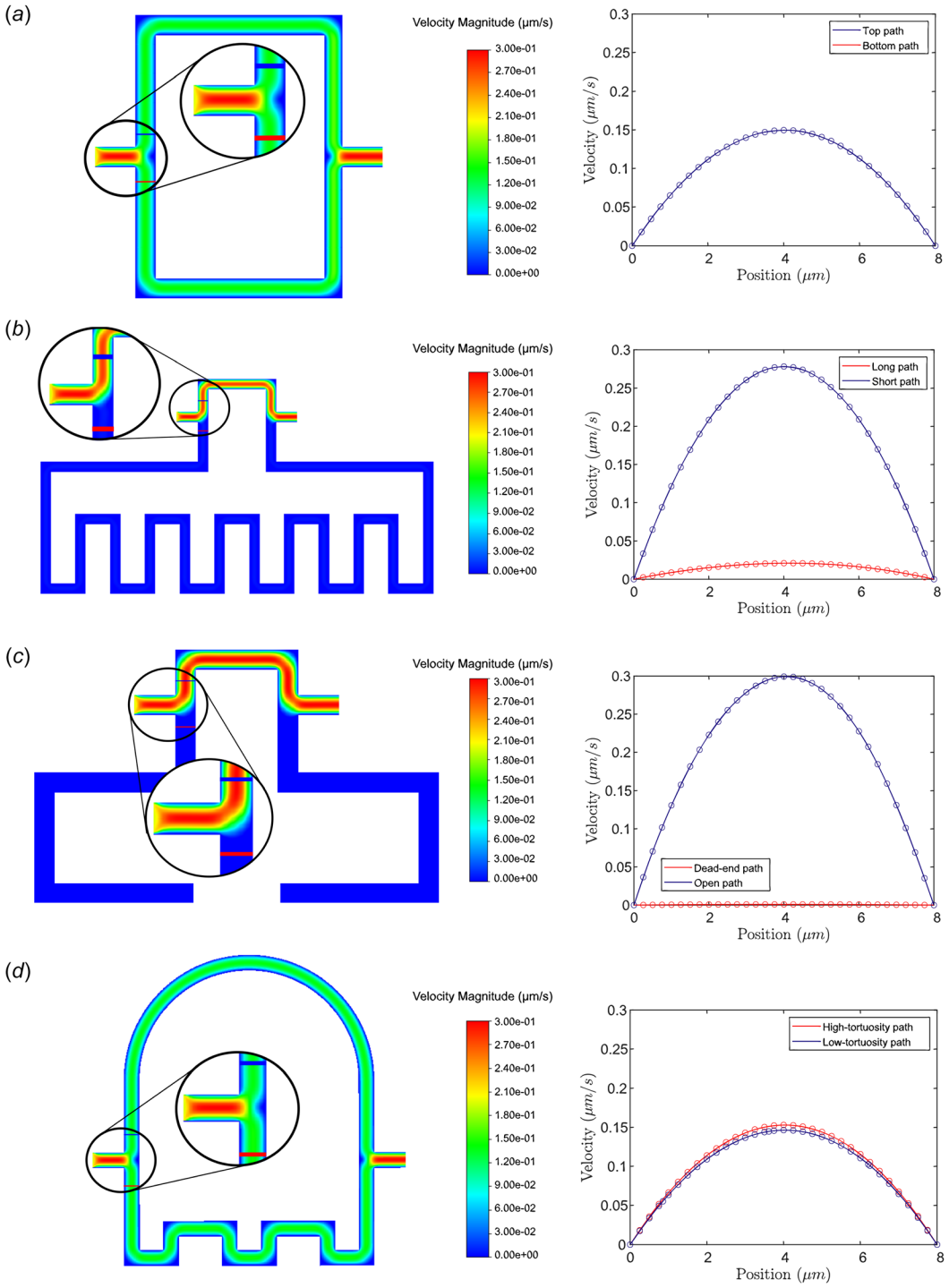


Fig. 3 Contours of the velocity magnitude (μm/s) (left) and the velocity profile (right) among the two paths of the symmetric microdevice (*a*), the twisted microdevice (*b*), the dead-end microdevice (*c*), and the tortuosity microdevice (*d*). The velocity profiles (right) of each microdevice correspond to the sections marked (top/blue and bottom/red) in the magnified detail of the contours (left).

pose higher hydraulic resistance since cells could not push the fluid any further from the wall. The results (Fig. 4) revealed that 77% of the cells chose the open path channel to migrate (devices 6, cells 13).

In addition, we sought to understand whether the tortuosity of the paths could also influence the decision-making process. For this reason, new migration channels were designed with similar lengths and hydraulic resistances but different degrees of tortuosity. As shown (Fig. 4), MDA-MD-231 cancer cells presented a higher predisposition to migrate through the high-tortuosity path (devices 4, cells 16).

A representative video of cell migration for each geometry is shown in Videos S2, S3, S4, and S5 available in the Supplemental Materials on the ASME Digital Collection.

MDA-MB-231 Metastatic Cells Migrate Faster and More Effectively Than MCF-7 Nonmetastatic Cells Within a Three-Dimensional Extracellular Matrix. To corroborate whether the migration capabilities exhibited within confined microchannels could be extrapolated to extracellular matrices, we studied MCF-7

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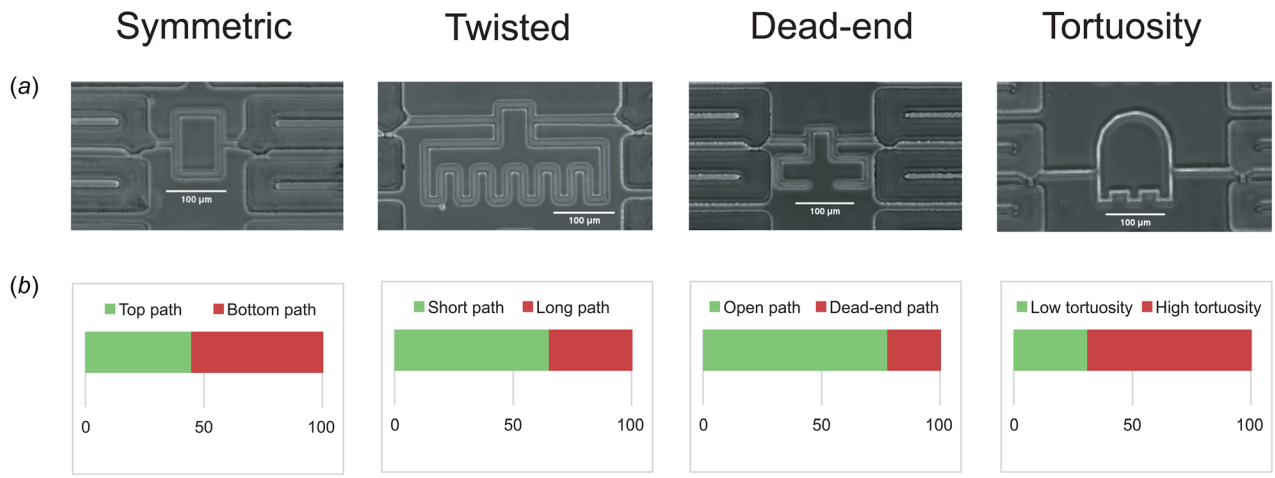


Fig. 4 Cell decision per device type. (*a*) Microscopy image of the internal geometry of each device; the scale bar represents 100 μm. (*b*) Results from cell decision in confined channels are expressed as the percentage of cells that migrate through each possible path. Each side/color represents a different migration path as previously shown (as a summary, left/green for upper paths and right/red for inferior paths). All experiments were conducted in at least four independent assays ($n = 4$), as shown in the text, with total cell counts per type of device being 18, 26, 13, and 16.

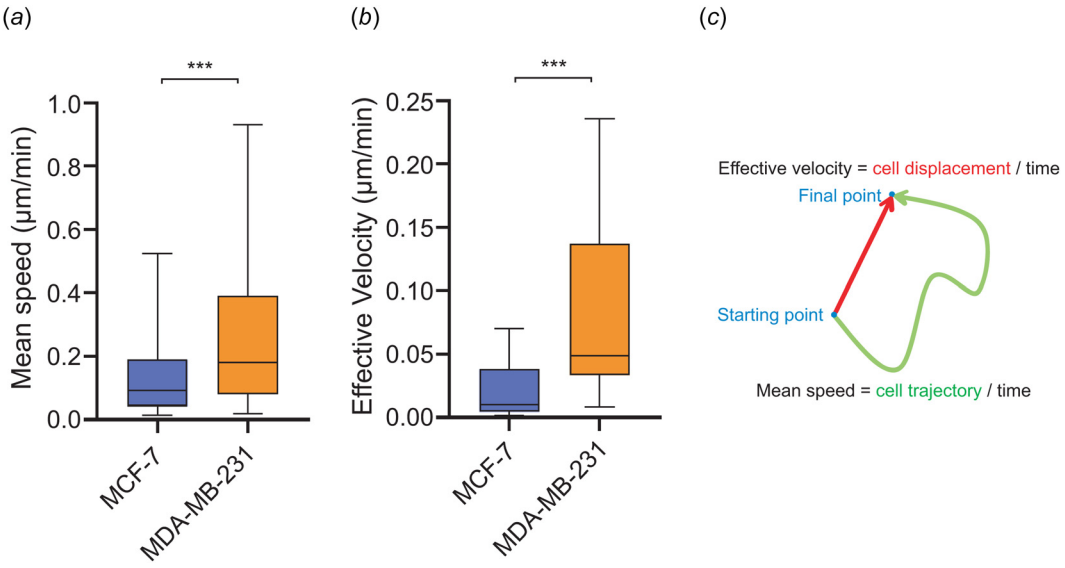


Fig. 5 (*a*) Mean speed and (*b*) effective velocity of MCF-7 and MDA-MB-231 cells in 2.5 mg/mL 3D collagen type I gels. (*c*) Outline indicating how the plotted parameters were calculated. Two independent devices per condition ($n = 2$) were used to study more than 90 cells and 6700 data points in total. Analysis of variance followed by posthoc Tukey–Kramer tests were performed to determine statistical significance: ***$p < 0.001$.

and MDA-MB-231 cell migration within 3D collagen type I for 24 h. Our results showed that MDA-MB-231 cells were significantly faster than MCF-7 cells (mean speed, Fig. 5(*a*)) and were much more effective in their migration (effective velocity, Fig. 5(*b*)).

Discussion

Here, we have presented new devices to trap single cancer cells and evaluate their behavior under confined paths and different hydraulic resistances. Our approach relies on recreating diverse restrictive 3D physiological environments, which are crucial aspects of tumor metastasis. Therefore, we have been able to determine the capacity of certain breast tumor cell lines (1) to get inside of, and (2) to migrate through confined spaces, as well as (3) to evaluate their migratory barotaxis-based decisions of metastatic cells. Replicating key steps for tumor cell spreading and metastasis formation will help to evaluate their metastatic potential and in turn, increase the number of drug targets as well as the chances of survival of patients.

Our results seem to exhibit a difference in constriction capabilities as well as migration behavior between cells with different metastatic potentials. Whereas 30% of metastatic MDA-MB-231 cells entered the confined channel (10 μm × 10 μm), a key step in the metastasis process [9], only 9% of nonmetastatic MCF-7 cells succeeded (nearly 1000 cells tested in total). A similar performance was observed in Yankaskas et al. [47], where 100-μm^2 channels (10 μm × 10 μm) allowed the entrance of 3% MCF-7 and 20% MDA-MB-231. Furthermore, low metastatic cancer cells (SKOV3) managed it with 5% of the cells in slightly smaller channels (60 μm^2: 10 μm × 6 μm) [31]. In addition, we quantified that while one-third of the MDA-MB-231 cells that entered the confined channel was able to reach the decision point, none of the MCF-7 cells achieved it.

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The behavior exposed by MCF-7 cells, typically characterized as nonmetastatic, paves the way for the implementation of such microdevices to determine the inherent migratory capability and to screen the metastatic potential of certain cancer cells. The results are in agreement with previously published work [48], which showed that under chemotactic stimulus with fetal bovine serum, MCF-7 cells were almost unable to enter and migrate through confined channels even of a 200-μm^2 ($10\,\mu m \times 20\,\mu m$) section. However, fetal bovine serum chemical stimulus was able to generate migration of MDA-MB-231 through channels that were nearly 10 times smaller [48]. This difference in migration abilities was also found within the 3D extracellular matrix (Fig. 5). MDA-MB-231 cells exhibited significantly faster and more directed migration than MCF-7 cells. Therefore, all these data seem to suggest both a reduced ability of MCF-7 cells to enter and migrate within the channels in comparison with MDA-MB-231 cells (as well as within 3D ECM). Furthermore, the previously observed spontaneous movement of confined cells [49] might not affect all cancer cells equally, considering the results found here.

The size of the microfluidic device channels was designed following Irimia recommendations for confined cancer cell migration [50], as well as other previously published works [48,49], in which a maximum migration speed was reported within channels of 75-μm^2 cross section [49]. The 80-μm^2 section channels ($10\,\mu m \times 8\,\mu m$) ensure that both MCF-7 and MDA-MB-231 cell sizes allow them to enter and be confined within these channels. The 19.39-μm ($\pm$2.82 μm SD.) and 16.78 μm ($\pm$2.39 μm) diameters found, respectively, in MCF-7 and MDA-MB-231 cells were similar to those of previous literature results [51,52]. In addition, given the ability of tumor epithelial cells to migrate in the absence of an extracellular matrix coating [49], we tested (results not shown) whether cells could enter the 80-μm^2 section channels. We found that neither MCF-7 nor MDA-MB-231 cells were able to squeeze into the channels. Thus, the channel entrance was slightly enlarged (up to 100 μm^2: $10\,\mu m \times 10\,\mu m$), creating a funnel-like shape that could facilitate the process. The cross-sectional area of the channel has always been very far from the 7 μm^2 ($2.65\,\mu m \times 2.65\,\mu m$) around which cell arrest is usually produced in tumor cells [53]. Nevertheless, no improvement was shown either with the enlarged section, suggesting that an external protein network is necessary for their entrance and consequent movement. Other strategies, such as culture medium pretreatment (alone) [32] or detergent pretreatment [54], were dismissed because of their inefficacy or interference with cell permeability.

Furthermore, taking advantage of the spontaneous movement of mechanically constrained cells [49,55], these devices present microchannels with a geometry that forces cells to make a decision on which path to migrate. Their choice is based exclusively on the hydraulic resistance perceived by the individual cells, since they are under confined conditions, mimicking biological preexistent structures and microchannel structures created by the ECM degradation of metalloproteinases [56]. The importance of such hydraulic resistance in cell mechanics, known as barotaxis, has been previously pointed out, showing that it can regulate polarization and cell motility [27]. However, whether this mechanism may drive metastasis in cancer or not is still an unsolved question.

In our model, metastatic breast cells preferentially migrated through fewer resistance paths. This directional bias becomes more evident as the hydraulic resistance difference increases. Our results agree with Prentice-Mott et al. [28], who showed that lymphocytes have an enhanced preference for the less resistant channel to migrate. Nevertheless, unlike lymphocytes, we observed that MDA-MB-231 cells also seem to be able to migrate through the more resistant channels in more than 20% of the cells. These results could be validated by other devices with comparable data [57,58], which suggested the use of the osmotic engine model in MDA-MB-231 cells to migrate through highly resistant channels. This mechanism, based on cell permeability and polarization, is able to generate a water flux through the cell and, in the end, cell displacement [25]. Then, although metastatic cancer cells seem to sense and tend to migrate through the lowest resistant channels, they may have underlying mechanisms, such as the osmotic engine model or their ability to squeeze [48], that allow them to overcome some difficulties encountered in the migration journey.

Moreover, with the aim of unraveling novel and unknown features influencing cancer cell migration, we built a tortuosity microdevice, whose bottom channel presents a higher level of tortuosity than the top channel, because of the number of bends that the path has compared to the top channel. In this configuration, the high-tortuosity path is slightly shorter than the low-tortuosity path to compensate for the hydraulic losses produced in the changes in the flow direction in elbow piping (available in the Supplemental Materials on the ASME Digital Collection). Thus, we achieved a microdevice with two channels with similar hydraulic resistances that permitted us to study whether the topology might affect the cell capacity to sense hydraulic resistances. Despite the fact that a light deviation in the cell migration toward the high-tortuosity path could be expected, the results show a marked preference in migrating through the high-tortuosity path than through the low-tortuosity path. These could be attributed to the slight differences in hydraulic resistance or the differential tortuosity of the channel, suggesting that still unknown mechanisms might play a part in cancer cell migration regarding the large variety of migration modes [9] as well as the still elusive interplay of the biophysical variables, such as confinement, topology, adhesion or rigidity [59].

Despite the many attempts to understand cancer metastasis and the increasing use of microfluidics [60–62], few devices have been designed and used to elucidate cancer cell migration in confined environments that specifically target barotaxis. Nevertheless, although not based on this type of taxis, they have already been proven to be a potential tool to predict the clinical outcomes and metastatic propensity of diverse tumors [47].

To our knowledge, this work originally presents a microfluidic-based study that integrates cancer cell migration through confined spaces and the analysis of metastasis capability based on a barotactic cell decision. However, future experiments will increase the number of cells and will diversify the metastatic potential of the cells tested to overcome the main limitations of this work.

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